

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	29 June 2026
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Priority
Sprint-1	Define Problem and Understanding	USN-1	As a researcher, I want to identify and define the agricultural problem that the crop recommendation system aims to solve.	High
Sprint-1		USN-2	As a stakeholder, I want to gather and analyze business requirements to understand project objectives and expected outcomes.	High
Sprint-1		USN-3	As a developer, I want to conduct a literature survey to study existing crop recommendation techniques and machine learning approaches.	Medium
Sprint-1		USN-4	As a policymaker, I want to analyze the social and business impact of implementing an AI-driven crop recommendation solution.	Medium
Sprint-1	Data Collection and Analysis	USN-5	As a data engineer, I want to download and collect the agricultural dataset from reliable sources.	High
Sprint-1		USN-6	As a developer, I want to import the required Python libraries for data analysis, visualization, and machine learning.	High
Sprint-1		USN-7	As a data analyst, I want to read and explore the dataset to understand its structure, features, and target variables.	High
Sprint-1		USN-8	As a data analyst, I want to perform univariate analysis to examine the distribution of N, P, K, temperature, humidity, pH, and rainfall.	Medium
Sprint-1		USN-9	As a data analyst, I want to conduct bivariate analysis to identify relationships between agricultural parameters and crop suitability.	Medium
Sprint-1		USN-10	As a data analyst, I want to perform multivariate analysis to discover patterns and interactions among multiple variables.	Low
Sprint-2	Data Pre-Processing	USN-11	As a data engineer, I want to check for null values and handle missing data to improve dataset quality.	High
Sprint-2		USN-12	As a data engineer, I want to detect and treat outliers that may affect model performance.	High
Sprint-2		USN-13	As a data engineer, I want to extract seasonal crop information and prepare relevant features for analysis.	Medium
Sprint-2		USN-14	As a data engineer, I want to split the dataset into training and testing sets for model development and evaluation.	High
Sprint-2	Model Building	USN-15	As a data scientist, I want to apply K-Means Clustering to identify patterns and group similar agricultural conditions.	High

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Priority
Sprint-2		USN-16	As a data scientist, I want to train a Logistic Regression model to predict suitable crops based on environmental parameters.	High
Sprint-2		USN-17	As a data scientist, I want to evaluate model performance using appropriate metrics and compare results.	High
Sprint-2		USN-18	As a data scientist, I want to save the best-performing model for deployment and generate crop predictions from user inputs.	High
Sprint-2	Application Building	USN-19	As a developer, I want to design and develop HTML pages to create an interactive and user-friendly interface.	High
Sprint-2		USN-20	As a developer, I want to build the Python backend code and integrate it with the trained machine learning model.	High
Sprint-2		USN-21	As a developer, I want to run, test, and validate the application to ensure accurate crop recommendations and smooth functionality.	High

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Sprint Release Date (Actual)
Sprint-1	23	7 Days	20 Jun 2025	26 Jun 2025	26 Jun 2025
Sprint-2	36	7 Days	27 Jun 2025	03 Jul 2025	03 Jul 2025

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\text{sprint duration}}{\text{velocity}} = \frac{20}{10} = 2$$

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

<https://www.visual-paradigm.com/scrum/scrum-burndown-chart/>

<https://www.atlassian.com/agile/tutorials/burndown-charts>

Reference:

<https://www.atlassian.com/agile/project-management>

<https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software>

<https://www.atlassian.com/agile/tutorials/epics>

<https://www.atlassian.com/agile/tutorials/sprints>

<https://www.atlassian.com/agile/project-management/estimation>

<https://www.atlassian.com/agile/tutorials/burndown-charts>